# RTI Display Controller

*Wiring Reference Arduino Mega 2560*

This document lists every physical connection needed for the Mega-based RTI display controller: the display itself, the open/close toggle switch, the ignition power relay, the rotary encoder, and the status LEDs. Pair this with the sketch `RTI_control_MEGA.ino` and the project README.

## 1. RTI Display

The RTI unit is receive-only - it has no data-out/TX line, so it can never send anything back to the Mega. The display communicates over serial at 2400 baud on the Mega's hardware Serial1 port, which keeps USB (Serial, pins 0/1) free for programming and debugging at the same time.

| RTI Display Pin | Signal | Mega Pin |
| --- | --- | --- |
| Pin 18 | RTI RX (data in) | 18 (TX1) |
| Pin 5 | GND | GND |

> Mega pin 19 (RX1) is not used for anything - the RTI unit has no TX line to receive from. Leave it disconnected.

## 2. Open / Close Toggle Switch

A simple SPDT (or SPST) toggle switch. One leg goes to a digital pin (configured with an internal pull-up), the other leg goes to GND. This switch no longer directly commands the display over serial - it now drives the relay in section 3, which physically cuts or restores the display's ignition power.

| Switch Leg | Mega Pin |
| --- | --- |
| Signal leg | 5 |
| Other leg | GND |

> Default assumption: pulling the pin LOW (switch closed to GND) means **open**. If your switch behaves backwards, see “Reversed behavior” at the end of this document.

## 3. Ignition Power Relay

Instead of trying to fake an “off” state over serial (unreliable), the display's actual +12V ignition wire is switched through a relay. Relay ON = ignition power reaches the display = open. Relay OFF = ignition cut = display fully unpowered = closed.

Use a proper relay module (the small board with a built-in transistor driver and flyback diode) - do not wire a bare relay coil directly to a Mega pin, it can't safely supply the current a coil needs.

| Connection | Goes To |
|---|---|
| Relay module signal/IN pin | Mega pin 11 |
| Relay module VCC | Mega 5V (or separate 5V supply if coil draws more current) |
| Relay module GND | Mega GND |
| Relay COM | +12V ignition wire |
| Relay NO (normally open) | Onward to RTI display's ignition input |

Wiring uses the relay's NO contacts specifically: de-energized (OFF) leaves the circuit open, so no ignition power reaches the display by default - a safer failure mode than using NC contacts. Most relay modules trigger ON with a HIGH signal, but some cheap active-low boards trigger ON with LOW instead; see “Reversed behavior” if yours is on when it should be off.

## 4. Rotary Encoder (with pushbutton)

A standard quadrature rotary encoder with an integrated pushbutton (e.g. KY-040 style). CLK must be wired to an interrupt-capable pin - pin 2 on the Mega is used here.

| Encoder Pin | Function | Mega Pin |
|---|---|---|
| CLK | Quadrature signal A (interrupt) | 2 |
| DT | Quadrature signal B | 3 |
| SW | Integrated pushbutton | 4 |
| GND | Ground | GND |
| + / VCC | 5V (if your encoder module needs it) | 5V |

Rotate clockwise to increase brightness, counter-clockwise to decrease. Press SW to cycle video mode (RGB → PAL → NTSC) while the display is open. If direction feels backwards, swap the CLK and DT wires - no code change needed.

## 5. Status LEDs

Five LEDs give visual feedback on the current state. Wire each through a 220-330Ω current-limiting resistor to GND (standard LED + resistor circuit, anode to the Mega pin, cathode through the resistor to GND).

| LED | Behavior | Mega Pin |
|---|---|---|
| Status | On when display is open, off when closed | 6 |
| Mode: RGB | On when RGB mode is active (and open) | 7 |
| Mode: PAL | On when PAL mode is active (and open) | 8 |
| Mode: NTSC | On when NTSC mode is active (and open) | 9 |
| Brightness | PWM - its own physical brightness mirrors the selected level (0-15 mapped to 0-255) | 10 |

## 6. Complete Pin Map

| Mega Pin | Connected To |
|---|---|
| 0 / 1 | Reserved - USB Serial (programming / Serial Monitor) |
| 2 | Encoder CLK |
| 3 | Encoder DT |
| 4 | Encoder SW (pushbutton) |
| 5 | Open/Close switch |
| 6 | Status LED |
| 7 | Mode LED - RGB |
| 8 | Mode LED - PAL |
| 9 | Mode LED - NTSC |
| 10 | Brightness LED (PWM) |
| 11 | Ignition relay signal/IN |
| 18 (TX1) | RTI display RX |

## 7. EEPROM Map (for reference, no wiring involved)

| Address | Stores |
|---|---|
| 0 | Current display mode |
| 1 | Current brightness level |
| 2 | “Enter” favorite brightness level |
| 3 | “Back” favorite brightness level |
| 4 | Last video mode used while open |

## 8. Reversed Behavior

These are the most likely “it works, just backwards” issues, and how to fix them without touching the wiring:

• **Switch is backwards** (closed lowers the display instead of raising it): in the sketch, find `checkOpenCloseSwitch()` and change `reading == LOW` to `reading == HIGH`.

• **Relay is backwards** (display is powered when it should be off, or vice versa): change `RELAY_ACTIVE_HIGH` from `true` to `false` near the top of the sketch.

• **Encoder direction is backwards** (clockwise dims instead of brightens): swap the CLK and DT wires at the encoder - no code change needed.

• **One encoder click changes brightness by more than one level**: change `ENCODER_STEPS_PER_CLICK` from 4 to 2 near the top of the sketch.

Note: these wiring assignments and behaviors are based on the sketch as written and standard component conventions - they haven't been verified against your specific physical parts. Confirm each connection against your actual hardware before powering everything on together.

# 8. Display connector pinout

| Control module terminal | Breakout box terminal | Signal type | |
|---|---|---|---|
| J1 | #1 | Color signal (red), screen | |
| J2 | #2 | Color signal (green), display | |
| J3 | #3 | Color signal (blue), screen | |
| J4 | #4 | Synchronizing signal, TV | |
| J5 | #5 | Screen ground, video signals | |
| J6 | #6 | Signal, display remote control | RX |
| J7 | #7 | - | |
| J8 | #8 | Screen ground, cable to the screen | RCA- |
| J9 | #9 | Video signal to display | RCA+ |
| J10 | #10 | - | |
| J11 | #11 | 30-supply (power supply from battery) | BAT+ |
| J12 | #12 | 30-supply (power supply from battery) | BAT+ |
| J13 | #13 | 15 supply | ACC (IGNITION) |
| J14 | #14 | - | |
| J15 | #15 | Ground (Measured to battery negative terminal) | BAT- |
| J16 | #16 | Ground (Measured to battery negative terminal) | BAT- |
| J17 | #17 | - | |
| J18 | #18 | Serial bus from the control module | TX |
| J19 | #19 | - | |
| J20 | #20 | - | |
| J21 | #21 | - | |
| J22 | #22 | - | |